

ASSIGNMENT:- 2

AIM:- Matrix Multiplication

PROBLEM STATEMENT:- Implement Matrix Multiplication using Map-Reduce

THOERY:-

Matrix multiplication is a fundamental operation in linear algebra, commonly used in scientific computing, data analysis, and machine learning. When dealing with large matrices, traditional approaches may not be efficient. To overcome scalability and performance issues, we can leverage Hadoop MapReduce, which allows distributed computation over large datasets.

What is Matrix Multiplication?

Matrix multiplication is an operation where two matrices are multiplied to produce a third matrix. It is defined only when the number of columns in the first matrix A is equal to the number of rows in the second matrix B .

Let:

- Matrix A be of size $m \times n$
- Matrix B be of size $n \times p$

Then, the resulting matrix $C = A \times B$ will be of size $m \times p$.

How Does It Work?

Each element in the resulting matrix $C[i][j]$ is calculated by taking the dot product of the i -th row of matrix A and the j -th column of matrix B .

$$C_{i,j} = \sum_{k=0}^{n-1} A_{i,k} \times B_{k,j}$$

MapReduce for Matrix Multiplication

1. Mapper Function

The Mapper reads each line of the input matrix and emits intermediate key-value pairs based on the rules of matrix multiplication.

- If the line is from matrix A , the mapper emits intermediate keys for every column in matrix B .
 - If the line is from matrix B , the mapper emits intermediate keys for every row in matrix A .
- Key: (i, j) for the resulting matrix
 - Value: A tuple indicating source matrix, column/row index, and value

2. Reducer Function

The Reducer receives all values for a given key (i,j). It:

- Separates values from matrix A and B ,

- Computes the dot product between the row of A and column of B by multiplying and summing up matching values.
- Output: (i, j, result) - which is the value at position (i, j) of the product matrix.

MapReduce Flow

- Input file is uploaded to HDFS.
- Mapper processes the file and emits intermediate key-value pairs.
- Hadoop shuffles and sorts intermediate data by key.
- Reducer takes each key, computes the final result, and outputs the product matrix elements.

CODE:-

Start Hadoop Services

1. Open terminal and switch to the Hadoop user

```
~$ su hduser
```

```
hduser@Ubuntu:/home/yash_zagekar$ cd
```

2. Start HDFS

```
hduser@Ubuntu:~$ start-dfs.sh
```

```
Starting namenodes on [localhost]
```

```
Starting datanodes
```

```
Starting secondary namenodes [Ubuntu]
```

```
2025-04-17 16:38:54,137 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
```

3. start-yarn.sh

```
hduser@Ubuntu:~$ start-yarn.sh
```

```
Starting resourcemanager
```

```
Starting nodemanagers
```

```
hduser@Ubuntu:~$ jps
```

```
5393 Jps
```

```
4579 DataNode
```

```
4820 SecondaryNameNode
```

```
4375 NameNode
```

```
hduser@Ubuntu:~$ ls
```

```
Desktop          mapper_word_count.py  snap
Documents        Music                 student_marks.csv
Downloads        Pictures              Templates
hadoop-3.3.4     Public               Videos
hadoop-3.3.4.tar.gz  reducer_student_grade.py word_count.txt
mapper_student_grade.py reducer_word_count.py
```

4. Create a text file with 2 matrices

```
hduser@Ubuntu:~$ nano matrix.txt
```

```
{
```

matrix.txt file will open in nano input file

```
}
```

```
hduser@Ubuntu:~$ cat matrix.txt
```

```
A,0,0,1
A,0,1,2
A,1,0,3
A,1,1,4
B,0,0,5
B,0,1,6
B,1,0,7
B,1,1,8
```

```
hduser@Ubuntu:~$ hadoop fs -ls /
```

```
2025-04-17 16:47:01,709 WARN util.NativeCodeLoader: Unable to load native-hadoop
library for your platform... using builtin-java classes where applicable
Found 4 items
drwxr-xr-x - hduser supergroup      0 2025-04-15 07:02 /mapreduce_student_grade
drwxr-xr-x - hduser supergroup      0 2025-04-14 17:34 /mapreduce_word_count
drwxr-xr-x - hduser supergroup      0 2025-04-14 18:00 /output
drwxr-xr-x - hduser supergroup      0 2025-04-15 07:32 /output_student_grade
```

5. Create a directory in HDFS and Upload your file

```
hduser@Ubuntu:~$ hadoop fs -mkdir /
```

```
mapreduce_matrix_multiplication
```

```
2025-04-17 16:47:30,213 WARN util.NativeCodeLoader: Unable to load native-hadoop
library for your platform... using builtin-java classes where applicable
```

```
hduser@Ubuntu:~$ hadoop fs -ls /
```

```
2025-04-17 16:47:39,252 WARN util.NativeCodeLoader: Unable to load native-hadoop
library for your platform... using builtin-java classes where applicable
Found 5 items
```

```
drwxr-xr-x - hduser supergroup      0 2025-04-17 16:47 /
```

```
mapreduce_matrix_multiplication
```

```
drwxr-xr-x - hduser supergroup      0 2025-04-15 07:02 /mapreduce_student_grade
drwxr-xr-x - hduser supergroup      0 2025-04-14 17:34 /mapreduce_word_count
drwxr-xr-x - hduser supergroup      0 2025-04-14 18:00 /output
drwxr-xr-x - hduser supergroup      0 2025-04-15 07:32 /output_student_grade
```

```
hduser@Ubuntu:~$ hadoop fs -put matrix.txt /
```

```
mapreduce_matrix_multiplication
```

```
2025-04-17 16:48:25,553 WARN util.NativeCodeLoader: Unable to load native-hadoop
library for your platform... using builtin-java classes where applicable
```

```
hduser@Ubuntu:~$ hadoop fs -ls /mapreduce_matrix_multiplication
```

```
2025-04-17 16:48:50,554 WARN util.NativeCodeLoader: Unable to load native-hadoop
library for your platform... using builtin-java classes where applicable
Found 1 items
```

```
-rw-r--r--  1 hduser supergroup      64 2025-04-17 16:48 /
```

```
mapreduce_matrix_multiplication/matrix.txt
```

```
hduser@Ubuntu:~$ hadoop fs -cat /
```

```
mapreduce_matrix_multiplication/matrix.txt
```

```
2025-04-17 16:49:24,121 WARN util.NativeCodeLoader: Unable to load native-hadoop
library for your platform... using builtin-java classes where applicable
```

```
A,0,0,1
A,0,1,2
A,1,0,3
A,1,1,4
B,0,0,5
B,0,1,6
B,1,0,7
B,1,1,8
```

6. Create Mapper and Reducer Python Scripts

```
hduser@Ubuntu:~$ nano mapper_matrix.py
{
    After this command mapper_matrix.py will open in nano
    Write the mapper code for mapper_matrix
}

hduser@Ubuntu:~$ nano reducer_matrix.py
{
    After this command reducer_matrix.py will open in nano
    Write the mapper code for reducer_matrix
}

hduser@Ubuntu:~$ chmod +x mapper_matrix.py

hduser@Ubuntu:~$ chmod +x reducer_matrix.py
```

7. Locate Hadoop Streaming JAR

Now open new terminal (keep the old terminal active) and note the path

```
~$ su hduser

hduser@Ubuntu:/home/yash_zagekar$ cd

hduser@Ubuntu:~$ cd /usr/local/hadoop/share/hadoop/tools/lib

hduser@Ubuntu:/usr/local/hadoop/share/hadoop/tools/lib$ ls
aliyun-java-sdk-core-4.5.10.jar      hadoop-gridmix-3.3.4.jar
aliyun-java-sdk-kms-2.11.0.jar      hadoop-kafka-3.3.4.jar
aliyun-java-sdk-ram-3.1.0.jar        hadoop-minicluster-3.3.4.jar
aliyun-sdk-oss-3.13.0.jar            hadoop-openstack-3.3.4.jar
aws-java-sdk-bundle-1.12.262.jar     hadoop-resourceestimator-3.3.4.jar
azure-data-lake-store-sdk-2.3.9.jar  hadoop-rumen-3.3.4.jar
azure-keyvault-core-1.0.0.jar        hadoop-sls-3.3.4.jar
azure-storage-7.0.1.jar              hadoop-streaming-3.3.4.jar
hadoop-aliyun-3.3.4.jar              hamcrest-core-1.3.jar
hadoop-archive-logs-3.3.4.jar        ini4j-0.5.4.jar
hadoop-archives-3.3.4.jar            jdk.tools-1.8.jar
hadoop-aws-3.3.4.jar                 jdom2-2.0.6.jar
hadoop-azure-3.3.4.jar               junit-4.13.2.jar
hadoop-azure-datalake-3.3.4.jar      kafka-clients-2.8.1.jar
hadoop-client-3.3.4.jar              lz4-java-1.7.1.jar
hadoop-datajoin-3.3.4.jar            ojalgo-43.0.jar
hadoop-distcp-3.3.4.jar              opentracing-api-0.33.0.jar
hadoop-dynamometer-blockgen-3.3.4.jar opentracing-noop-0.33.0.jar
hadoop-dynamometer-infra-3.3.4.jar   opentracing-util-0.33.0.jar
hadoop-dynamometer-workload-3.3.4.jar org.jacoco.agent-0.8.5-runtime.jar
hadoop-extras-3.3.4.jar             wildfly-openssl-1.0.7.Final.jar
hadoop-fs2img-3.3.4.jar              zstd-jni-1.4.9-1.jar

hduser@Ubuntu:/usr/local/hadoop/share/hadoop/tools/lib$ pwd
/usr/local/hadoop/share/hadoop/tools/lib
```

8. Go back to your old terminal!

Paste the directory got from new terminal

`/usr/local/hadoop/share/hadoop/tools/lib/hadoop-streaming-3.3.4.jar`

```
hduser@Ubuntu:~$ hadoop jar /usr/local/hadoop/share/hadoop/
tools/lib/hadoop-streaming-3.3.4.jar -input /
mapreduce_matrix_multiplication/matrix.txt -output /
output_matrix -mapper /home/hduser/mapper_matrix.py -reducer /
home/hduser/reducer_matrix.py
```

****Make sure you correctly give your own saved txt and py file names.**

```
2025-04-17 17:04:09,289 WARN util.NativeCodeLoader: Unable to load native-hadoop
library for your platform... using builtin-java classes where applicable
2025-04-17 17:04:10,415 INFO impl.MetricsConfig: Loaded properties from hadoop-
metrics2.properties
2025-04-17 17:04:10,764 INFO impl.MetricsSystemImpl: Scheduled Metric snapshot period
at 10 second(s).
2025-04-17 17:04:10,765 INFO impl.MetricsSystemImpl: JobTracker metrics system started
2025-04-17 17:04:10,814 WARN impl.MetricsSystemImpl: JobTracker metrics system already
initialized!
2025-04-17 17:04:11,221 INFO mapred.FileInputFormat: Total input files to process : 1
2025-04-17 17:04:11,384 INFO mapreduce.JobSubmitter: number of splits:1
2025-04-17 17:04:11,676 INFO mapreduce.JobSubmitter: Submitting tokens for job:
job_local1421426574_0001
2025-04-17 17:04:11,676 INFO mapreduce.JobSubmitter: Executing with tokens: []
2025-04-17 17:04:12,147 INFO mapreduce.Job: The url to track the job: http://
localhost:8080/
2025-04-17 17:04:12,159 INFO mapreduce.Job: Running job: job_local1421426574_0001
2025-04-17 17:04:12,188 INFO mapred.LocalJobRunner: OutputCommitter set in config null
2025-04-17 17:04:12,191 INFO mapred.LocalJobRunner: OutputCommitter is
org.apache.hadoop.mapred.FileOutputCommitter
2025-04-17 17:04:12,223 INFO output.FileOutputCommitter: File Output Committer
Algorithm version is 2
2025-04-17 17:04:12,223 INFO output.FileOutputCommitter: FileOutputCommitter skip
cleanup _temporary folders under output directory:false, ignore cleanup failures:
false
2025-04-17 17:04:12,427 INFO mapred.LocalJobRunner: Waiting for map tasks
2025-04-17 17:04:12,437 INFO mapred.LocalJobRunner: Starting task:
attempt_local1421426574_0001_m_000000_0
2025-04-17 17:04:12,518 INFO output.FileOutputCommitter: File Output Committer
Algorithm version is 2
2025-04-17 17:04:12,518 INFO output.FileOutputCommitter: FileOutputCommitter skip
cleanup _temporary folders under output directory:false, ignore cleanup failures:
false
2025-04-17 17:04:12,597 INFO mapred.Task: Using ResourceCalculatorProcessTree : [ ]
2025-04-17 17:04:12,622 INFO mapred.MapTask: Processing split: hdfs://localhost:54310/
mapreduce_matrix_multiplication/matrix.txt:0+64
2025-04-17 17:04:12,677 INFO mapred.MapTask: numReduceTasks: 1
2025-04-17 17:04:12,812 INFO mapred.MapTask: (EQUATOR) 0 kvi 26214396(104857584)
2025-04-17 17:04:12,812 INFO mapred.MapTask: mapreduce.task.io.sort.mb: 100
2025-04-17 17:04:12,812 INFO mapred.MapTask: soft limit at 83886080
2025-04-17 17:04:12,812 INFO mapred.MapTask: bufstart = 0; bufvoid = 104857600
2025-04-17 17:04:12,812 INFO mapred.MapTask: kvstart = 26214396; length = 6553600
2025-04-17 17:04:12,816 INFO mapred.MapTask: Map output collector class =
org.apache.hadoop.mapred.MapTask$MapOutputBuffer
2025-04-17 17:04:12,818 INFO streaming.PipeMapRed: PipeMapRed exec [/home/hduser/
mapper_matrix.py]
2025-04-17 17:04:12,826 INFO Configuration.deprecation: mapred.work.output.dir is
deprecated. Instead, use mapreduce.task.output.dir
2025-04-17 17:04:12,827 INFO Configuration.deprecation: mapred.local.dir is
deprecated. Instead, use mapreduce.cluster.local.dir
2025-04-17 17:04:12,828 INFO Configuration.deprecation: map.input.file is deprecated.
Instead, use mapreduce.map.input.file
2025-04-17 17:04:12,828 INFO Configuration.deprecation: map.input.length is
deprecated. Instead, use mapreduce.map.input.length
2025-04-17 17:04:12,828 INFO Configuration.deprecation: mapred.job.id is deprecated.
Instead, use mapreduce.job.id
```

```

2025-04-17 17:04:12,829 INFO Configuration.deprecation: mapred.task.partition is
deprecated. Instead, use mapreduce.task.partition
2025-04-17 17:04:12,832 INFO Configuration.deprecation: map.input.start is deprecated.
Instead, use mapreduce.map.input.start
2025-04-17 17:04:12,834 INFO Configuration.deprecation: mapred.task.is.map is
deprecated. Instead, use mapreduce.task.ismap
2025-04-17 17:04:12,835 INFO Configuration.deprecation: mapred.task.id is deprecated.
Instead, use mapreduce.task.attempt.id
2025-04-17 17:04:12,835 INFO Configuration.deprecation: mapred.tip.id is deprecated.
Instead, use mapreduce.task.id
2025-04-17 17:04:12,835 INFO Configuration.deprecation: mapred.skip.on is deprecated.
Instead, use mapreduce.job.skiprecords
2025-04-17 17:04:12,841 INFO Configuration.deprecation: user.name is deprecated.
Instead, use mapreduce.job.user.name
2025-04-17 17:04:13,050 INFO streaming.PipeMapRed: R/W/S=1/0/0 in:NA [rec/s] out:NA
[rec/s]
2025-04-17 17:04:13,058 INFO streaming.PipeMapRed: Records R/W=8/1
2025-04-17 17:04:13,061 INFO streaming.PipeMapRed: MRErrorThread done
2025-04-17 17:04:13,062 INFO streaming.PipeMapRed: mapRedFinished
2025-04-17 17:04:13,072 INFO mapred.LocalJobRunner:
2025-04-17 17:04:13,075 INFO mapred.MapTask: Starting flush of map output
2025-04-17 17:04:13,076 INFO mapred.MapTask: Spilling map output
2025-04-17 17:04:13,076 INFO mapred.MapTask: bufstart = 0; bufend = 192; bufvoid =
104857600
2025-04-17 17:04:13,076 INFO mapred.MapTask: kvstart = 26214396(104857584); kvend =
26214336(104857344); length = 61/6553600
2025-04-17 17:04:13,117 INFO mapred.MapTask: Finished spill 0
2025-04-17 17:04:13,149 INFO mapred.Task: Task:attempt_local1421426574_0001_m_000000_0
is done. And is in the process of committing
2025-04-17 17:04:13,158 INFO mapred.LocalJobRunner: Records R/W=8/1
2025-04-17 17:04:13,160 INFO mapred.Task: Task
'attempt_local1421426574_0001_m_000000_0' done.
2025-04-17 17:04:13,182 INFO mapred.Task: Final Counters for
attempt_local1421426574_0001_m_000000_0: Counters: 23

```

File System Counters

```

FILE: Number of bytes read=141410
FILE: Number of bytes written=786978
FILE: Number of read operations=0
FILE: Number of large read operations=0
FILE: Number of write operations=0
HDFS: Number of bytes read=64
HDFS: Number of bytes written=0
HDFS: Number of read operations=5
HDFS: Number of large read operations=0
HDFS: Number of write operations=1
HDFS: Number of bytes read erasure-coded=0

```

Map-Reduce Framework

```

Map input records=8
Map output records=16
Map output bytes=192
Map output materialized bytes=230
Input split bytes=117
Combine input records=0
Spilled Records=16
Failed Shuffles=0
Merged Map outputs=0
GC time elapsed (ms)=0
Total committed heap usage (bytes)=168820736

```

File Input Format Counters

```

Bytes Read=64

```

```

2025-04-17 17:04:13,183 INFO mapred.LocalJobRunner: Finishing task:
attempt_local1421426574_0001_m_000000_0

```

2025-04-17 17:04:13,184 INFO mapred.LocalJobRunner: map task executor complete.
2025-04-17 17:04:13,187 INFO mapred.LocalJobRunner: Waiting for reduce tasks
2025-04-17 17:04:13,187 INFO mapred.LocalJobRunner: Starting task:
attempt_local1421426574_0001_r_000000_0
2025-04-17 17:04:13,196 INFO output.FileOutputCommitter: File Output Committer
Algorithm version is 2
2025-04-17 17:04:13,196 INFO output.FileOutputCommitter: FileOutputCommitter skip
cleanup _temporary folders under output directory:false, ignore cleanup failures:
false
2025-04-17 17:04:13,196 INFO mapred.Task: Using ResourceCalculatorProcessTree : []
2025-04-17 17:04:13,202 INFO mapred.ReduceTask: Using ShuffleConsumerPlugin:
org.apache.hadoop.mapreduce.task.reduce.Shuffle@310aalb6
2025-04-17 17:04:13,206 WARN impl.MetricsSystemImpl: JobTracker metrics system already
initialized!
2025-04-17 17:04:13,216 INFO mapreduce.Job: Job job_local1421426574_0001 running in
uber mode : false
2025-04-17 17:04:13,218 INFO mapreduce.Job: map 100% reduce 0%
2025-04-17 17:04:13,247 INFO reduce.MergeManagerImpl: MergerManager:
memoryLimit=546098368, maxSingleShuffleLimit=136524592, mergeThreshold=360424928,
ioSortFactor=10, memToMemMergeOutputsThreshold=10
2025-04-17 17:04:13,252 INFO reduce.EventFetcher:
attempt_local1421426574_0001_r_000000_0 Thread started: EventFetcher for fetching Map
Completion Events
2025-04-17 17:04:13,310 INFO reduce.LocalFetcher: localfetcher#1 about to shuffle
output of map attempt_local1421426574_0001_m_000000_0 decomp: 226 len: 230 to MEMORY
2025-04-17 17:04:13,318 INFO reduce.InMemoryMapOutput: Read 226 bytes from map-output
for attempt_local1421426574_0001_m_000000_0
2025-04-17 17:04:13,327 INFO reduce.MergeManagerImpl: closeInMemoryFile -> map-output
of size: 226, inMemoryMapOutputs.size() -> 1, commitMemory -> 0, usedMemory -> 226
2025-04-17 17:04:13,333 INFO reduce.EventFetcher: EventFetcher is interrupted..
Returning
2025-04-17 17:04:13,338 INFO mapred.LocalJobRunner: 1 / 1 copied.
2025-04-17 17:04:13,340 INFO reduce.MergeManagerImpl: finalMerge called with 1 in-
memory map-outputs and 0 on-disk map-outputs
2025-04-17 17:04:13,368 INFO mapred.Merger: Merging 1 sorted segments
2025-04-17 17:04:13,368 INFO mapred.Merger: Down to the last merge-pass, with 1
segments left of total size: 220 bytes
2025-04-17 17:04:13,379 INFO reduce.MergeManagerImpl: Merged 1 segments, 226 bytes to
disk to satisfy reduce memory limit
2025-04-17 17:04:13,381 INFO reduce.MergeManagerImpl: Merging 1 files, 230 bytes from
disk
2025-04-17 17:04:13,382 INFO reduce.MergeManagerImpl: Merging 0 segments, 0 bytes from
memory into reduce
2025-04-17 17:04:13,382 INFO mapred.Merger: Merging 1 sorted segments
2025-04-17 17:04:13,382 INFO mapred.Merger: Down to the last merge-pass, with 1
segments left of total size: 220 bytes
2025-04-17 17:04:13,383 INFO mapred.LocalJobRunner: 1 / 1 copied.
2025-04-17 17:04:13,386 INFO streaming.PipeMapRed: PipeMapRed exec [/home/hduser/
reducer_matrix.py]
2025-04-17 17:04:13,397 INFO Configuration.deprecation: mapred.job.tracker is
deprecated. Instead, use mapreduce.jobtracker.address
2025-04-17 17:04:13,399 INFO Configuration.deprecation: mapred.map.tasks is
deprecated. Instead, use mapreduce.job.maps
2025-04-17 17:04:13,505 INFO streaming.PipeMapRed: R/W/S=1/0/0 in:NA [rec/s] out:NA
[rec/s]
2025-04-17 17:04:13,508 INFO streaming.PipeMapRed: R/W/S=10/0/0 in:NA [rec/s] out:NA
[rec/s]
2025-04-17 17:04:13,513 INFO streaming.PipeMapRed: Records R/W=16/1
2025-04-17 17:04:13,516 INFO streaming.PipeMapRed: MRErrorThread done
2025-04-17 17:04:13,517 INFO streaming.PipeMapRed: mapRedFinished
2025-04-17 17:04:13,629 INFO mapred.Task: Task:attempt_local1421426574_0001_r_000000_0
is done. And is in the process of committing
2025-04-17 17:04:13,637 INFO mapred.LocalJobRunner: 1 / 1 copied.
2025-04-17 17:04:13,637 INFO mapred.Task: Task attempt_local1421426574_0001_r_000000_0
is allowed to commit now
2025-04-17 17:04:13,705 INFO output.FileOutputCommitter: Saved output of task
'attempt_local1421426574_0001_r_000000_0' to hdfs://localhost:54310/output_matrix
2025-04-17 17:04:13,709 INFO mapred.LocalJobRunner: Records R/W=16/1 > reduce
2025-04-17 17:04:13,709 INFO mapred.Task: Task
'attempt_local1421426574_0001_r_000000_0' done.
2025-04-17 17:04:13,710 INFO mapred.Task: Final Counters for
attempt_local1421426574_0001_r_000000_0: **Counters: 30**

File System Counters

FILE: Number of bytes read=141902
FILE: Number of bytes written=787208
FILE: Number of read operations=0
FILE: Number of large read operations=0
FILE: Number of write operations=0
HDFS: Number of bytes read=64
HDFS: Number of bytes written=36
HDFS: Number of read operations=10
HDFS: Number of large read operations=0
HDFS: Number of write operations=3
HDFS: Number of bytes read erasure-coded=0

Map-Reduce Framework

Combine input records=0
Combine output records=0
Reduce input groups=4
Reduce shuffle bytes=230
Reduce input records=16
Reduce output records=4
Spilled Records=16
Shuffled Maps =1
Failed Shuffles=0
Merged Map outputs=1
GC time elapsed (ms)=0
Total committed heap usage (bytes)=168820736

Shuffle Errors

BAD_ID=0
CONNECTION=0
IO_ERROR=0
WRONG_LENGTH=0
WRONG_MAP=0
WRONG_REDUCE=0

File Output Format Counters

Bytes Written=36

2025-04-17 17:04:13,710 INFO mapred.LocalJobRunner: Finishing task:
attempt_local1421426574_0001_r_000000_0
2025-04-17 17:04:13,710 INFO mapred.LocalJobRunner: reduce task executor complete.
2025-04-17 17:04:14,222 INFO mapreduce.Job: map 100% reduce 100%
2025-04-17 17:04:14,223 INFO mapreduce.Job: Job job_local1421426574_0001 completed
successfully

2025-04-17 17:04:14,253 INFO mapreduce.Job: **Counters: 36**

File System Counters

FILE: Number of bytes read=283312
FILE: Number of bytes written=1574186
FILE: Number of read operations=0
FILE: Number of large read operations=0
FILE: Number of write operations=0
HDFS: Number of bytes read=128
HDFS: Number of bytes written=36
HDFS: Number of read operations=15
HDFS: Number of large read operations=0
HDFS: Number of write operations=4
HDFS: Number of bytes read erasure-coded=0


```

Map-Reduce Framework
Map input records=8
Map output records=16
Map output bytes=192
Map output materialized bytes=230
Input split bytes=117
Combine input records=0
Combine output records=0
Reduce input groups=4
Reduce shuffle bytes=230
Reduce input records=16
Reduce output records=4
Spilled Records=32
Shuffled Maps =1
Failed Shuffles=0
Merged Map outputs=1
GC time elapsed (ms)=0
Total committed heap usage (bytes)=337641472
Shuffle Errors
BAD_ID=0
CONNECTION=0
IO_ERROR=0
WRONG_LENGTH=0
WRONG_MAP=0
WRONG_REDUCE=0
File Input Format Counters
Bytes Read=64
File Output Format Counters
Bytes Written=36
2025-04-17 17:04:14,254 INFO streaming.StreamJob: Output
directory: /output_matrix

```

9. View Output

```

hduser@Ubuntu:~$ hadoop fs -ls /output_matrix
2025-04-17 17:11:31,287 WARN util.NativeCodeLoader: Unable to load native-hadoop
library for your platform... using builtin-java classes where applicable
Found 2 items
-rw-r--r--  1 hduser supergroup          0 2025-04-17 17:04 /output_matrix/
_SUCCESS
-rw-r--r--  1 hduser supergroup        36 2025-04-17 17:04 /output_matrix/
part-00000

```

```

hduser@Ubuntu:~$ hadoop fs -cat /output_matrix/part-00000
2025-04-17 17:11:45,160 WARN util.NativeCodeLoader: Unable to load native-hadoop
library for your platform... using builtin-java classes where applicable
0,0  19.0
0,1  22.0
1,0  43.0
1,1  50.0

```

CONCLUSION:-

Matrix multiplication using Hadoop MapReduce is a classic example of dividing a computational task across nodes in a distributed system. This approach showcases the scalability and efficiency of Hadoop when dealing with large matrix data, and demonstrates how traditional algorithms can be restructured for distributed frameworks.